higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 4

(12010484)

1 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL drawings must be neat and well balanced.
5. ALL work you do not want to be marked must be clearly crossed out.
6. Write neatly and legibly.

NCV

QUESTION 1: PLASTICS

1.1 Give TWO examples of the following polymers:

1.1.1 Natural polymers

1.1.2 Synthetic polymers

(2 × 2) (4)

1.2 Name FIVE advantages of using plastic in the construction industry. (5)

1.3. State SIX uses of glass–reinforced plastic (GRP). (6)

1.4 Explain how plastic pipes are joined by using the solvent-welding method. (5)

[20]

QUESTION 2: ELASTOMERS

2.1 Name THREE properties of elastomers used in the building industry because of their desirable properties. (3)

2.2 State TWO applications (uses) of elastomers in the construction industry. (2)

2.3 Explain the procedure that you will follow when applying a silicone sealant to masonry and concrete. (5)

[10]

QUESTION 3: ADHESIVES

3.1 Why is it important to prepare (clean) the surface before you can apply wood glue? (3)

3.2 Name TWO types of PVA wood glue. (2)

3.3 Name FIVE advantages of using PVA emulsion wood glues. (5)

3.4 Name FIVE uses of polyurethane wood glue. (5)

[15]

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QUESTION 4: PAINTS, WATER REPELLENT, COATINGS AND PRESERVATIVES

4.1 Paint can harm your health because it contains chemical compounds that are toxic.

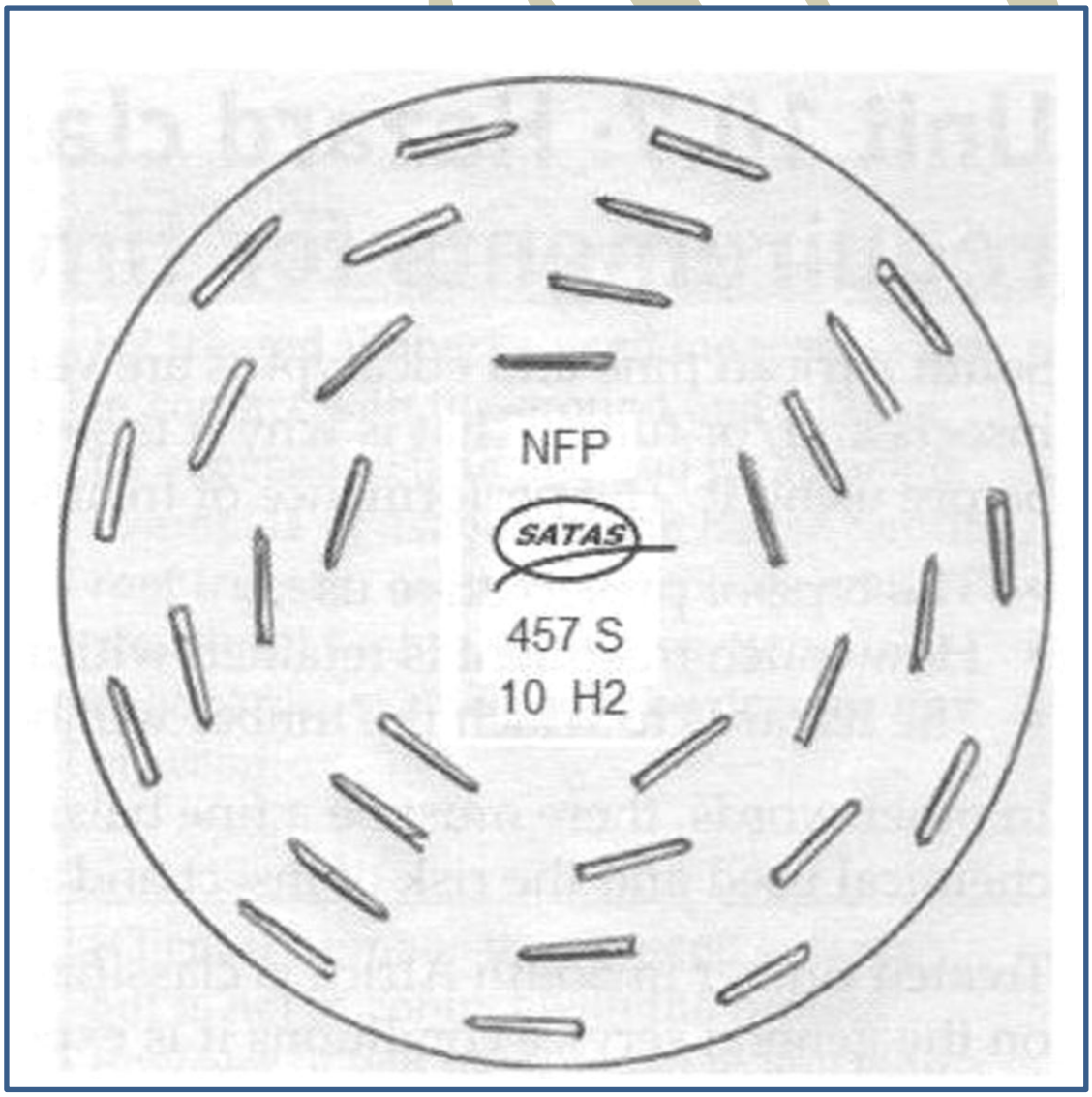
What general safety precautions will you take when painting? (5)

4.2 You are required to paint an object made from non-ferrous metal.

Explain the basic cleaning procedure you will use to prepare the object for painting. (5)

4.3 The diagram below indicates the logo of the South African technical auditing services (SATAS) certification mark on treated poles.

Identify the markings on the treated pole and explain what each marking on the plate represent.



(5)

4.4 Name FIVE advantages of wet blasting. (5)

[20]

QUESTION 5: WATERPROOFING AND DAMP-PROOFING AND VAPOUR BARRIER INSTALLATION

5.1 Name the step-by-step procedure on how to apply torch-on (polymer-modified bitumen membranes) to a roof. (5)

5.2 Sketch, (not to scale) a section of a face brick with a bullnose sill externally for brick wall (steel window) showing the following:

One brick wall plastered internally
Bull-nose external brick sill
Design of your DPC
Terrazzo internal tile (10)
[15]

QUESTION 6: BITUMEN, TAR AND ASPHALT

6.1 The employer must ensure that appropriate PPC (personal protective clothing) and PPE (personal protective equipment) are provided to workers.

Name FIVE of these PPCs that should be given to employees. (5)

6.2 Name TWO environmental hazards of tar. (2)

6.3 Name THREE tests that are carried out on bitumen. (3)
[10]

QUESTION 7: FASTENERS

7.1 Fasteners can be identified in terms of its head style or shape. Identify the following screws and write the name of the screw next to the corresponding number (7.1.1–7.1.4) in the ANSWER BOOK.

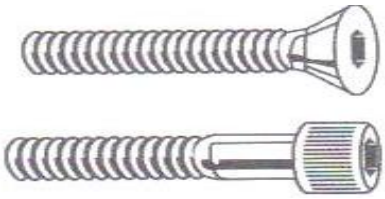
7.1.1



7.1.2



7.1.3



7.1.4



(4 × 1) (4)

7.2 Name FOUR functions of the bolt list. (4)

7.3 Name TWO failures that can still occur when tightening bolts in a structure connection. (2)

[10]

TOTAL: 100

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